

# WindOps Copilot - Turbine Report

Turbine ID: WTG-001 | Correlation ID: corr-16a4e640783b  
Report Generated: 2026-06-26T10:13:43.316579

TURBINE HEALTH STATUS

50.0% Health

## Live Sensor Telemetry

| Sensor Parameter      | Value  | Unit |
|-----------------------|--------|------|
| Wind Speed            | 11.80  | m/s  |
| Rotor Speed           | 16.29  | RPM  |
| Blade Temperature     | 24.28  | °C   |
| Generator Temperature | 74.83  | °C   |
| Vibration             | 0.27   | mm/s |
| Power Output          | 2.11   | MW   |
| Operational Status    | active |      |

## Active Fault Alerts

| Parameter      | Value | Threshold | Severity |
|----------------|-------|-----------|----------|
| Vibration      | 0.27  | 0.25      | CRITICAL |
| Generator Temp | 74.83 | 65.00     | WARNING  |

## Diagnostic Root Cause Analysis

|                     |                                                                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Diagnosed Cause:    | Generator Bearing Degraded / Gearbox Defect                                                                                                                                    |
| Confidence Level:   | 85.0%                                                                                                                                                                          |
| Technical Analysis: | Vibrations are high (0.27 mm/s) accompanied by elevated generator temperatures (74.8°C). This indicates friction, misalignment, or bearing breakdown in the drive train/shaft. |

## Operational Action Recommendations

| Recommended Action | Description | Risk | Production Loss | Confidence |
|--------------------|-------------|------|-----------------|------------|
|--------------------|-------------|------|-----------------|------------|

|                                     |                                                                                                                                                        |      |          |      |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|------|
| Emergency<br>Controlled<br>Shutdown | Trigger pitch feathers and rotor brake immediately due to critical vibration threshold breach (>0.25 mm/s). Protect structural integrity of the asset. | LOW  | severe   | 100% |
| De-rate / Curtail<br>Power Output   | Reduce generator speed. (NOT RECOMMENDED: safety thresholds are violated, structural damage is likely if kept spinning).                               | HIGH | moderate | 10%  |

*Note: This report is generated by an autonomous multi-agent system. final actions must be reviewed and authorized by a human operator.*